Cpts 317. 8/26.

you shall start to do your hw.

Automata	Languages.
‖	‖
special programs	sets of words.

how "special" about RU programs/automata?

$\Sigma = \{a, b\}$ is the alphabet. Consider

$\mathcal{L}_1 = \{w : w \text{ contains at least one } b\}$.

Example w in $\mathcal{L}_1$:

$w = aaaaababa$

need only finite memory.

(So, should't declare any int-var).

day 1:

day 2:

day 3:

⋮

day 9:

$\mathcal{L}_2 = \{ w : \text{in } w,$ ① a's are before b's
② $\#_a(w) = \#_b(w) \}$.

Example w in $\mathcal{L}_2$:

$$w = \overbrace{aaaaa}^{\leftarrow n \rightarrow}\overbrace{bbbbb}^{\leftarrow n \rightarrow}$$

day1:

day2:

⋮

day5:

day6:

⋮

day10:

you must(?) use memory to count $\#_a(w) = n$. How many bits needed to store n?

$n = 1024 \Rightarrow$ you need $\log_2 n = 10$ bits to store n.

as $n \to \infty$, you need inf. memory!

Operations on languages.

1. Union.

$$\mathcal{L}_1 \cup \mathcal{L}_2 := \{ w : w \in \mathcal{L}_1 \text{ or } w \in \mathcal{L}_2 \}.$$

Example: $\mathcal{L}_1 = \{a, ab\}$, $\mathcal{L}_2 = \{b, ab\}$,

$$\mathcal{L}_1 \cup \mathcal{L}_2 = \{a, b, ab\}.$$

(We also write $\mathcal{L}_1 + \mathcal{L}_2$ for $\mathcal{L}_1 \cup \mathcal{L}_2$).

2. Concatenation.

$$\mathcal{L}_1 \cdot \mathcal{L}_2 = \{ xy : x \in \mathcal{L}_1 \text{ and } y \in \mathcal{L}_2 \}.$$

(also we simply write $\mathcal{L}_1\mathcal{L}_2$ for $\mathcal{L}_1 \cdot \mathcal{L}_2$).

Exaple. $\mathcal{L}_1 = \{a, ba\}$, $\mathcal{L}_2 = \{b, ba\}$.

Then,

$$\mathcal{L}_1\mathcal{L}_2 = \{ab, aba, bab, baba\}$$

Exmple: $\mathcal{L}_1 = \emptyset$, $\mathcal{L}_2 = \{bb, ba\}$.

$$\mathcal{L}_1 \mathcal{L}_2 = \emptyset.$$

Define $L^k = \underbrace{L \cdot L \cdot L \cdot \ldots \cdot L}_{k \text{ times}}$.

Exmple: $L = \{a, ba\}$.

$abaa \in L^3$? yes

$abba \in L^3$? no.

$L^0 = \{\Lambda\}$ even when $L = \emptyset$.

$L^1 = L$

$L^2 = L \cdot L$

$L^3 = L \cdot L \cdot L$

$\vdots$